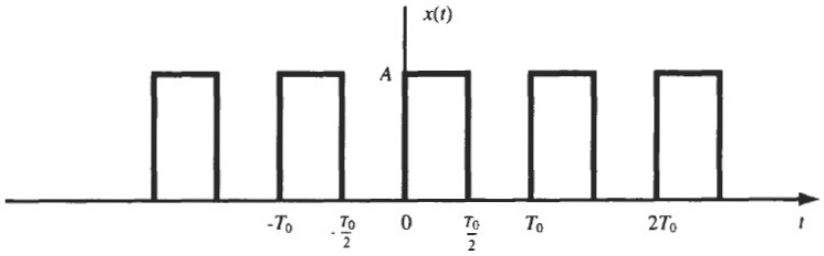
1) 주기 신호 x(t)가 다음 그림과 같을 때, 복수지수 형 푸리에 계수들, X_k, X_0을 구하시오. (답만 있으면 0점 처리함, 수식 전개과정 쓸 것 $e^{-j\pi} = -1$)

a)



$$x(t) = \begin{cases} A & for \quad 0 \le t \le \frac{1}{2}T_0 \\ 0 & \text{for} \quad \frac{1}{2}T_0 \le t \le T_0 \end{cases}$$

$$X_k = \frac{1}{T_0}\int_0^{\frac{1}{2}T_0} (A)e^{-j(2\pi/T_0)kt}\,dt = \left(\frac{A}{T_0}\right)\frac{e^{-j(2\pi/T_0)kt}}{-j(2\pi/T_0)k}\Bigg|_0^{T_0/2}$$

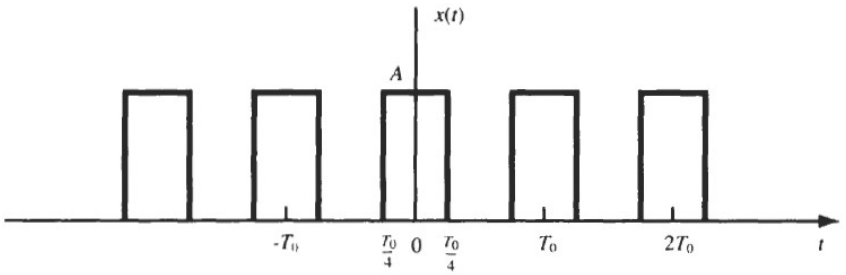
$$= \left(\frac{A}{T_0}\right)\frac{e^{-j(2\pi/T_0)k(1/2)T_0} - e^{-j(2\pi/T_0)k(0)}}{-j(2\pi/T_0)k} = (A)\frac{e^{-j\pi k} - 1}{-j2\pi k}$$

- 여기서 $e^{-j\pi} = -1$ 이므로 $X_k = (A)\dfrac{1-(-1)^k}{j2\pi k} \quad for \quad k \neq 0$

$$X_0 = \frac{1}{T_0}\int_0^{\frac{1}{2}T_0} (A)e^{-j0t}\,dt = \frac{A}{T_0}\left(\frac{1}{2}T_0\right) = \frac{A}{2}$$

$$X_k = \begin{cases} \dfrac{A}{j\pi k} & k = \pm 1, \pm 3, \pm 5, \cdots \\ 0 & k = \pm 2, \pm 4, \pm 6, \cdots \\ \dfrac{A}{2} & k = 0 \end{cases}$$

b)



$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T_0}$$

Using Eq. (*5.102b*), we have

$$c_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T_0}\int_{-T_0/4}^{T_0/4} A\, e^{-jk\omega_0 t}\, dt$$

$$= \frac{A}{-jk\omega_0 T_0}\left(e^{-jk\omega_0 T_0/4} - e^{jk\omega_0 T_0/4}\right)$$

$$= \frac{A}{-jk2\pi}\left(e^{-jk\pi/2} - e^{jk\pi/2}\right) = \frac{A}{k\pi}\sin\left(\frac{k\pi}{2}\right)$$

Thus,

$$c_k = 0 \qquad k = 2m \neq 0$$

$$c_k = (-1)^m \frac{A}{k\pi} \qquad k = 2m + 1$$

$$c_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\, dt = \frac{1}{T_0}\int_0^{T_0/2} A\, dt = \frac{A}{2}$$

Hence,

$$c_0 = \frac{A}{2} \qquad c_{2m} = 0,\ m \neq 0 \qquad c_{2m+1} = (-1)^m \frac{A}{(2m+1)\pi}$$